Assignment No. 9

1. Create an index that will enable a user to pull orders for '1990-10-03' out of the Orders table quickly.

```
KD3_rutuja_96226>CREATE INDEX idx_orders_odate ON orders(odate)
    -> ;
Query OK, 0 rows affected, 1 warning (2.119 sec)
Records: 0  Duplicates: 0  Warnings: 1

KD3_rutuja_96226>select * from orders;
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3001 |   18.69 | 1990-10-03 | 2008 | 1007 |
| 3003 |  767.19 | 1990-10-03 | 2001 | 1001 |
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3005 | 5160.45 | 1990-10-03 | 2003 | 1002 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3007 |   75.75 | 1990-10-04 | 2004 | 1002 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3010 |  309.95 | 1990-10-04 | 2004 | 1002 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+------+---------+------------+------+------+
10 rows in set (0.246 sec)
```

2. If the Orders table has already been created, how can you force the onum field to be unique (assume all current values are unique)?

```
KD3_rutuja_96226>ALTER table orders add constraint uniq_onum UNIQUE(onum);
Query OK, 0 rows affected (0.387 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

```
KD3_rutuja_96226>SHOW INDEX FROM orders;
+--------+------------+-----------------+--------------+-------------+-----------+-------------
+----------+--------+-----+------------+---------+---------------+---------+------------+
| Table  | Non_unique | Key_name        | Seq_in_index | Column_name | Collation | Cardinality
| Sub_part | Packed | Null | Index_type | Comment | Index_comment | Visible | Expression |
+--------+------------+-----------------+--------------+-------------+-----------+-------------
+----------+--------+-----+------------+---------+---------------+---------+------------+
| orders |          0 | uniq_onum       |            1 | onum        | A         |          10
|     NULL |   NULL | YES  | BTREE      |         |               | YES     | NULL       |
| orders |          1 | index1          |            1 | odate       | A         |           2
|     NULL |   NULL | YES  | BTREE      |         |               | YES     | NULL       |
| orders |          1 | idx_orders_odate |           1 | odate       | A         |           2
|     NULL |   NULL | YES  | BTREE      |         |               | YES     | NULL       |
+--------+------------+-----------------+--------------+-------------+-----------+-------------
+----------+--------+-----+------------+---------+---------------+---------+------------+
3 rows in set (0.990 sec)
```

3. Create an index that would permit salesperson to retrieve his orders.

```
KD3_rutuja_96226>CREATE index index_order_snum on orders(snum);
Query OK, 0 rows affected (0.632 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

4. Let us assume that each salesperson is to have only one customer of a given rating, and this is currently the case. Create an index that enforces it.

```
KD3_rutuja_96226>CREATE UNIQUE INDEX idx_salesperson_rating1
    -> ON customers (snum, rating);
ERROR 1062 (23000): Duplicate entry '1001-100' for key 'customers.idx_salesperson_rating1'
```

5. Create an index to speed up searching order on a given date by given customer.

```
KD3_rutuja_96226>CREATE INDEX idx_order_odate_cnum on orders(odate, cnum);
Query OK, 0 rows affected (0.493 sec)
Records: 0  Duplicates: 0  Warnings: 0
```